

Smart Student ERP System

Complete System Design Documentation



Project	Smart Student ERP System
Version	1.0.0
Diagrams	18 Complete Diagrams
Categories	UML, ER, DFD, Flowchart
Technology	Java Backend + PostgreSQL + HTML/CSS/JS
Academic Year	2024-25

Table of Contents

#	Diagram Type	Category	Page
1	Class Diagram	UML Structural	3
2	Object Diagram	UML Structural	4
3	Component Diagram	UML Structural	5
4	Deployment Diagram	UML Structural	6
5	Package Diagram	UML Structural	7
6	Composite Structure Diagram	UML Structural	8
7	Profile Diagram	UML Structural	9
8	Use Case Diagram	UML Behavioral	10
9	Activity Diagram	UML Behavioral	11
10	State Machine Diagram	UML Behavioral	12

11	ER Diagram	Database	13
12	EER Diagram	Database	14
13	Relational Schema	Database	15
14	Data Flow Diagram L0	DFD	16
15	Data Flow Diagram L1	DFD	17
16	Flowchart — Login	Flowchart	18
17	Flowchart — Fee Payment	Flowchart	19
18	Workflow Diagram	Workflow	20

1. Class Diagram

UML Structural Diagram — Smart Student ERP System

A Class Diagram shows the static structure of the system — classes, their attributes, methods, and relationships (inheritance, association, aggregation, composition).

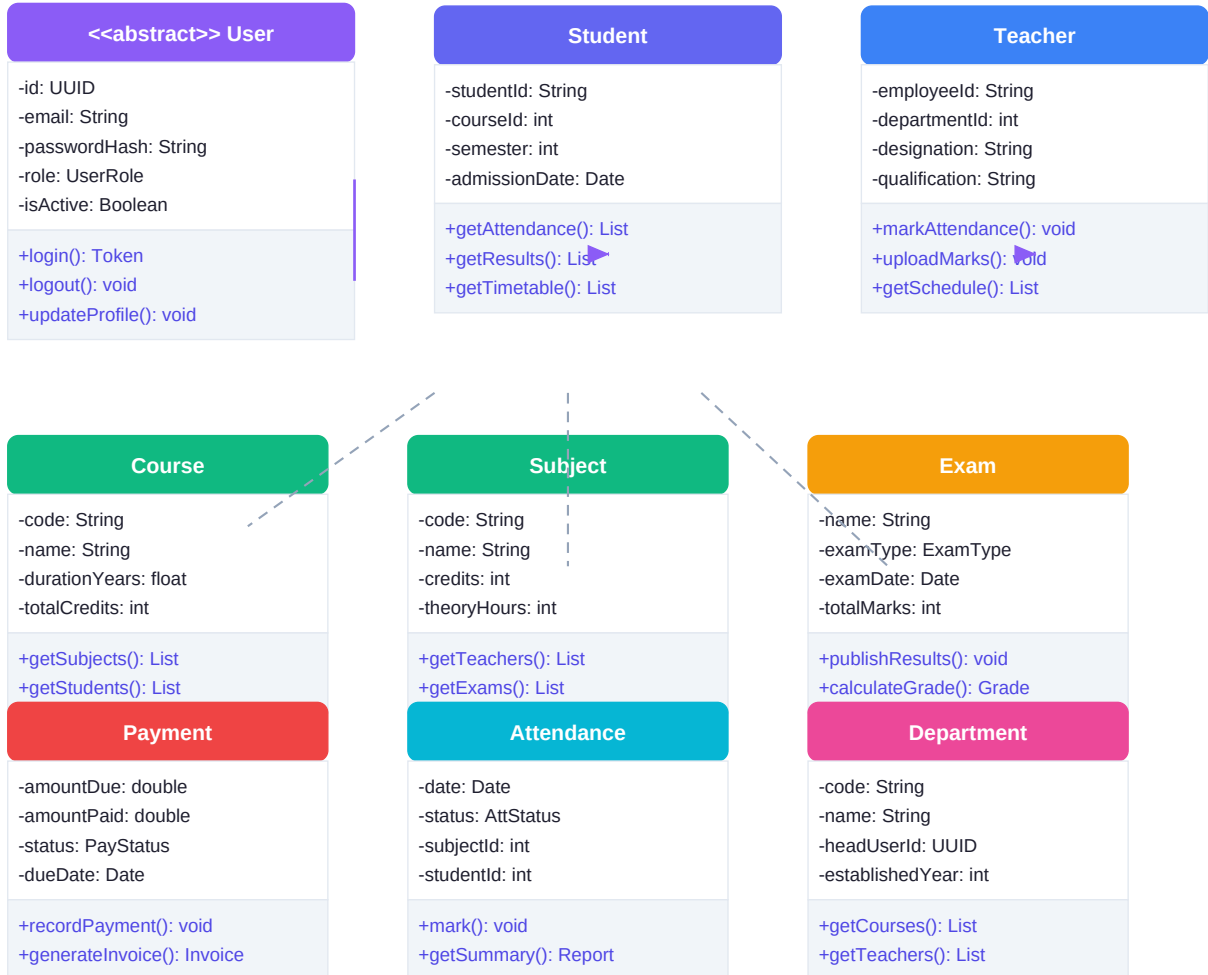


Figure 1: Class Diagram showing all major classes, attributes, methods and relationships

Legend:

Symbol	Meaning
—■	Inheritance (IS-A relationship)
--■	Dependency / Association
◆—	Composition (strong ownership)
■—	Aggregation (weak ownership)
+method()	Public method
-attribute	Private attribute

2. Object Diagram

UML Structural — Snapshot of system objects at runtime

An Object Diagram shows a specific snapshot of the system at a point in time — actual instances of classes with real values.

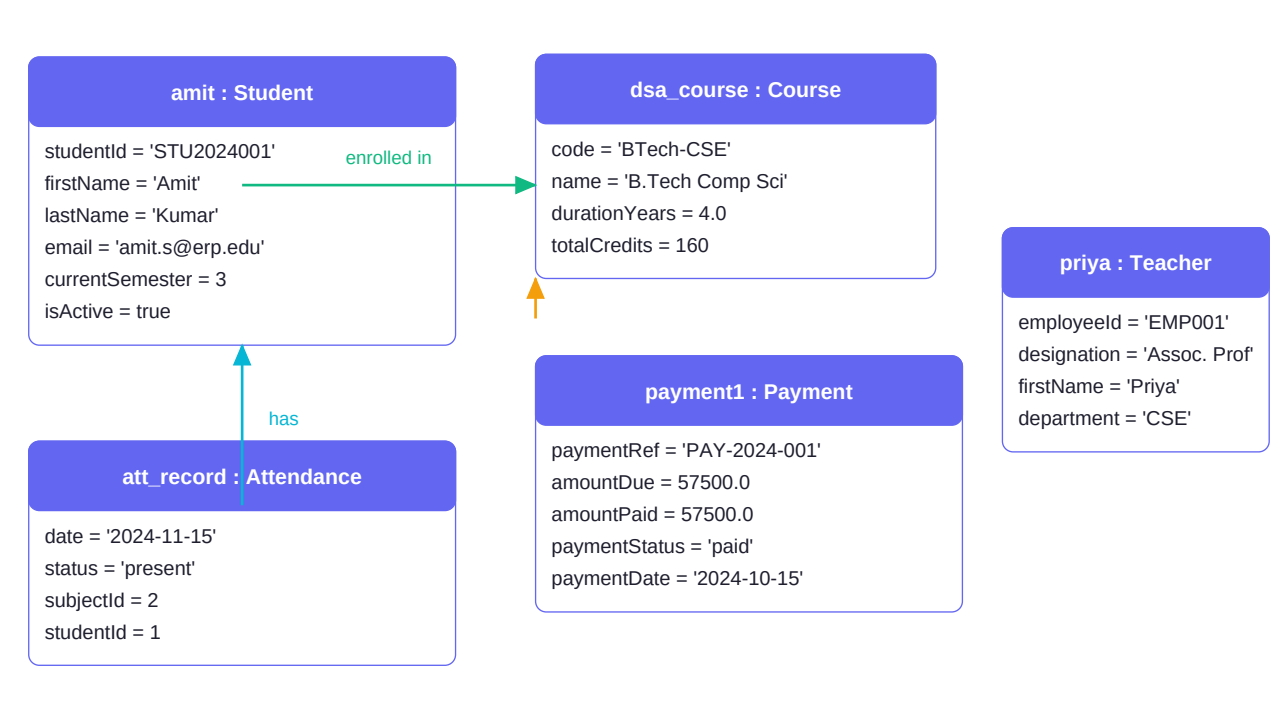


Figure 2: Object Diagram — runtime snapshot showing student Amit enrolled in BTech-CSE

Key observations: Objects show actual data values. Links between objects represent actual associations at runtime. This snapshot shows the state during Semester 3 of academic year 2024-25.

3. Component Diagram

UML Structural — System component architecture

Shows how the system is divided into components and the dependencies and interfaces between them.

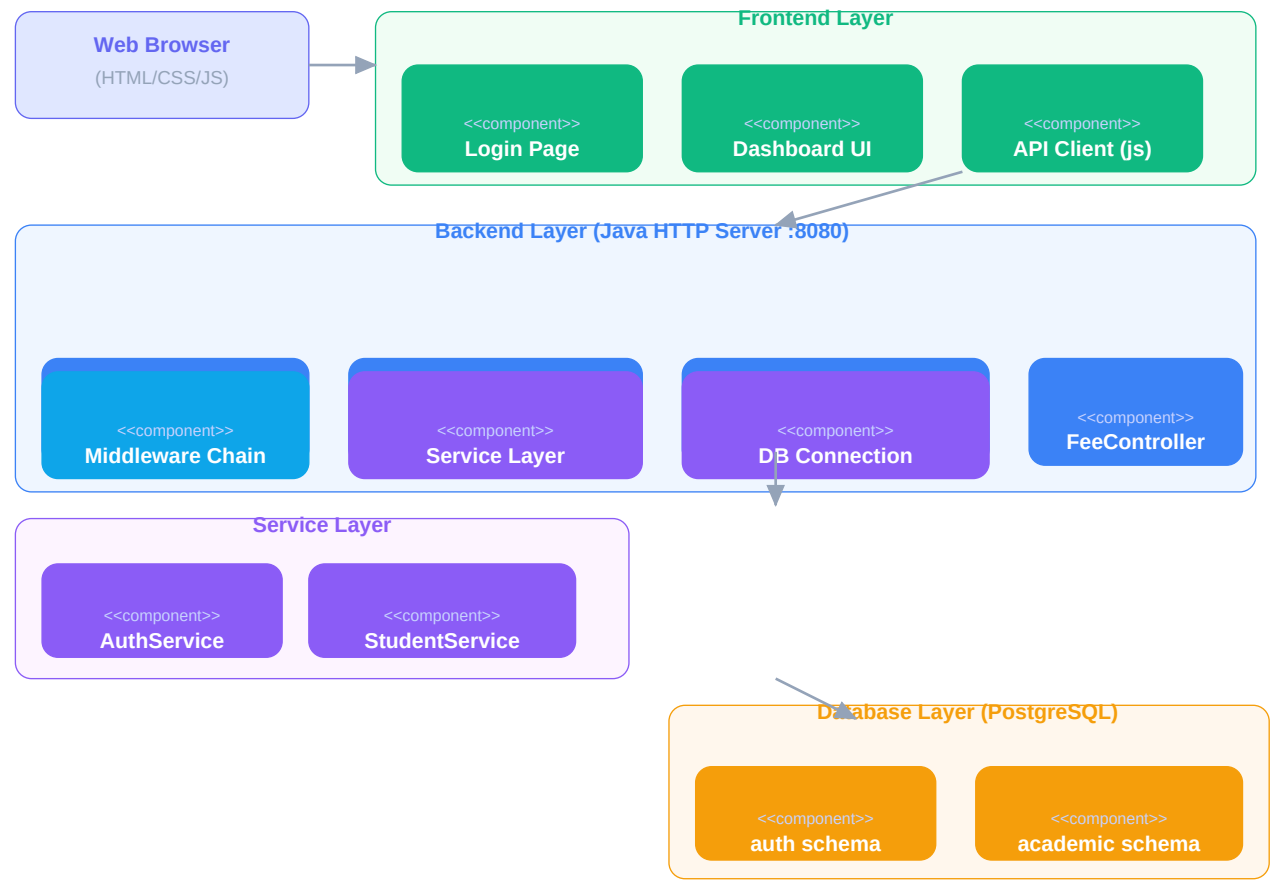


Figure 3: Component Diagram showing frontend, backend, service, and database layers

Component	Technology	Responsibility
Web Browser	HTML5/CSS3/JS	User interface rendering
Frontend Layer	Vanilla JavaScript	API calls, UI logic, routing
AuthController	Java	Login, logout, OTP, sessions
StudentController	Java	Student CRUD, profile management
AnalyticsController	Java	Dashboard statistics and reports
Service Layer	Java	Business logic, data processing
DB Connection	JDBC/PostgreSQL	Singleton connection pool
PostgreSQL	PostgreSQL 13+	Persistent data storage

4. Deployment Diagram

UML Structural — Physical deployment of system nodes

Shows how software components are deployed on hardware nodes (servers, devices) and the communication paths between them.

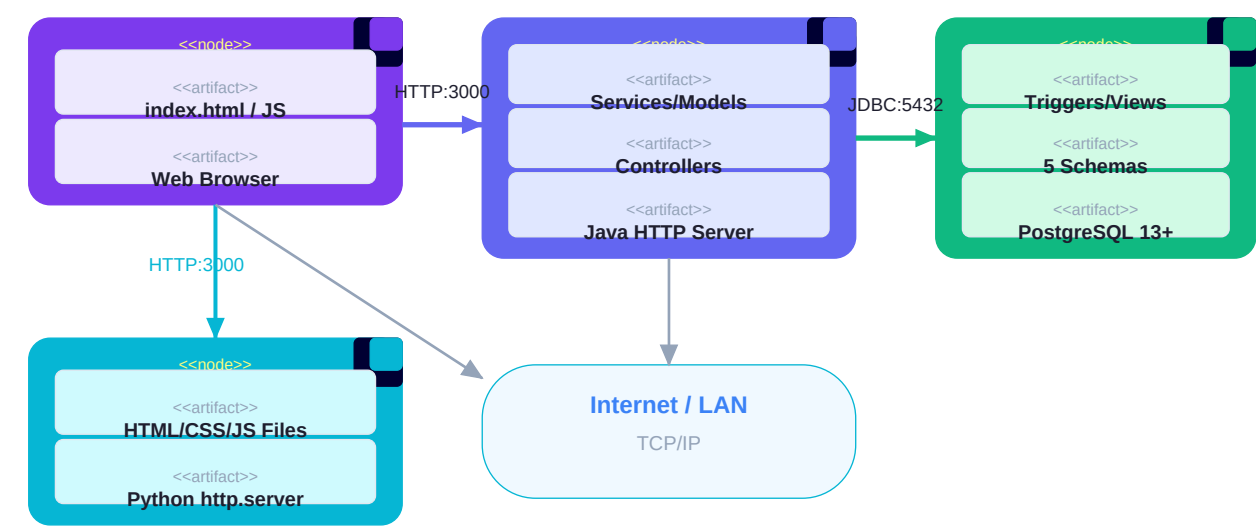


Figure 4: Deployment Diagram — local development setup with 3 servers

Node	Host	Port	Technology	Purpose
Client Device	localhost	—	Web Browser	User accesses the ERP UI
Static File Server	localhost	3000	Python HTTP Server	Serves HTML/CSS/JS files
Application Server	localhost	8080	Java HTTP Server	REST API backend
Database Server	localhost	5432	PostgreSQL 13+	Persistent data storage

5. Package Diagram

UML Structural — Package organization and dependencies

Shows how classes are organized into packages (namespaces) and the dependencies between packages.

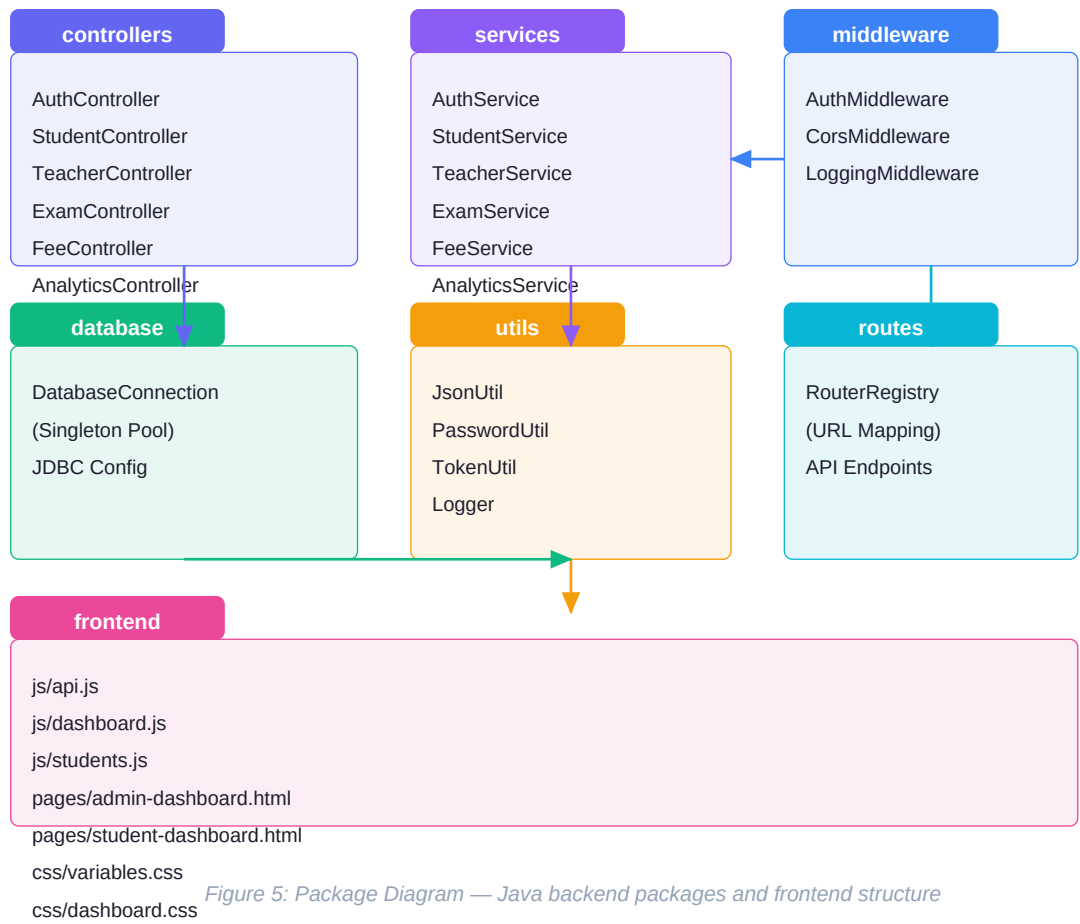


Figure 5: Package Diagram — Java backend packages and frontend structure

Package	Classes	Responsibility
controllers	13 classes	Handle HTTP requests, parse input, send responses
services	11 classes	Business logic, database operations, data processing
middleware	3 classes	Auth validation, CORS, request logging
database	1 class	Singleton PostgreSQL connection management
utils	4 classes	JSON parsing, password hashing, token generation, logging
routes	1 class	Central URL to controller mapping registry
frontend	20+ files	HTML pages, CSS styles, JavaScript modules

6. Composite Structure Diagram

UML Structural — Internal structure of AuthMiddleware

Shows the internal structure of a classifier (class/component), including its parts, ports, and connectors.

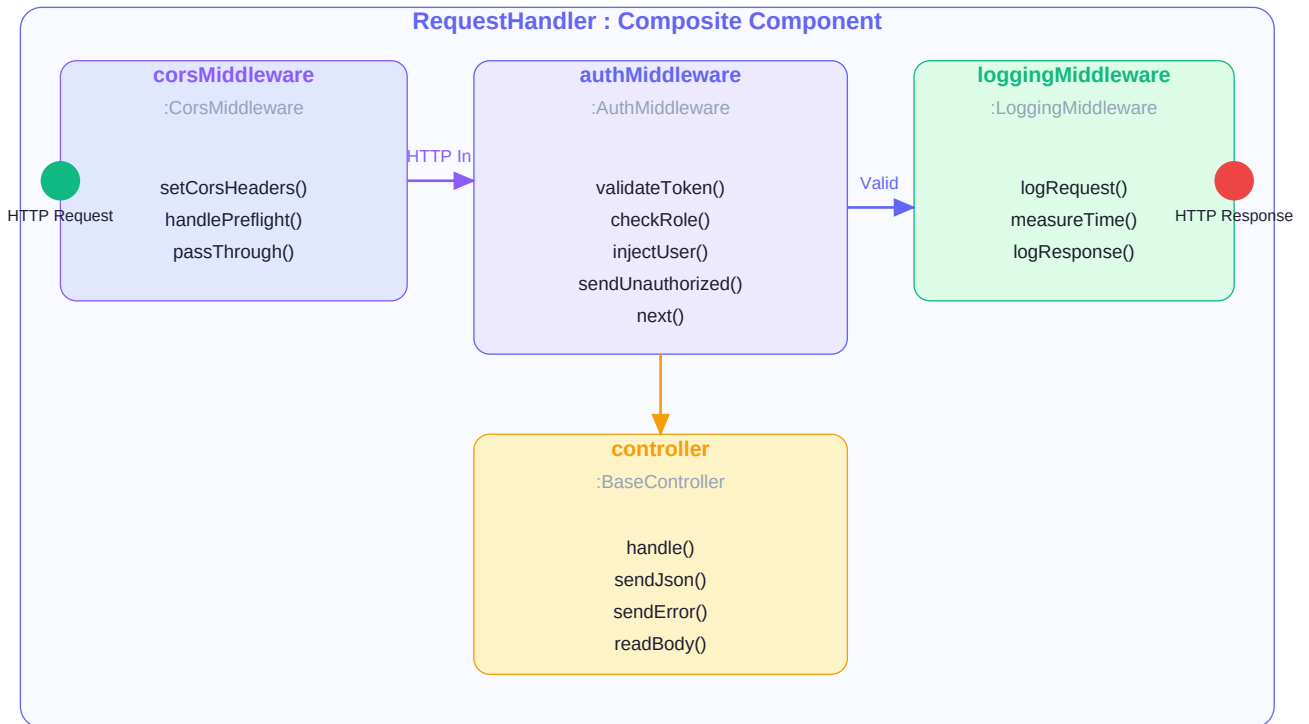


Figure 6: Composite Structure showing middleware chain internal parts and connectors

The RequestHandler composite processes every HTTP request through: (1) CorsMiddleware sets headers, (2) AuthMiddleware validates JWT token and injects user, (3) LoggingMiddleware records timing, (4) Controller handles the actual business logic.

7. Profile Diagram

UML Extension — Custom stereotypes for ERP domain

A Profile Diagram extends UML with domain-specific stereotypes, tagged values, and constraints for the Smart ERP system.

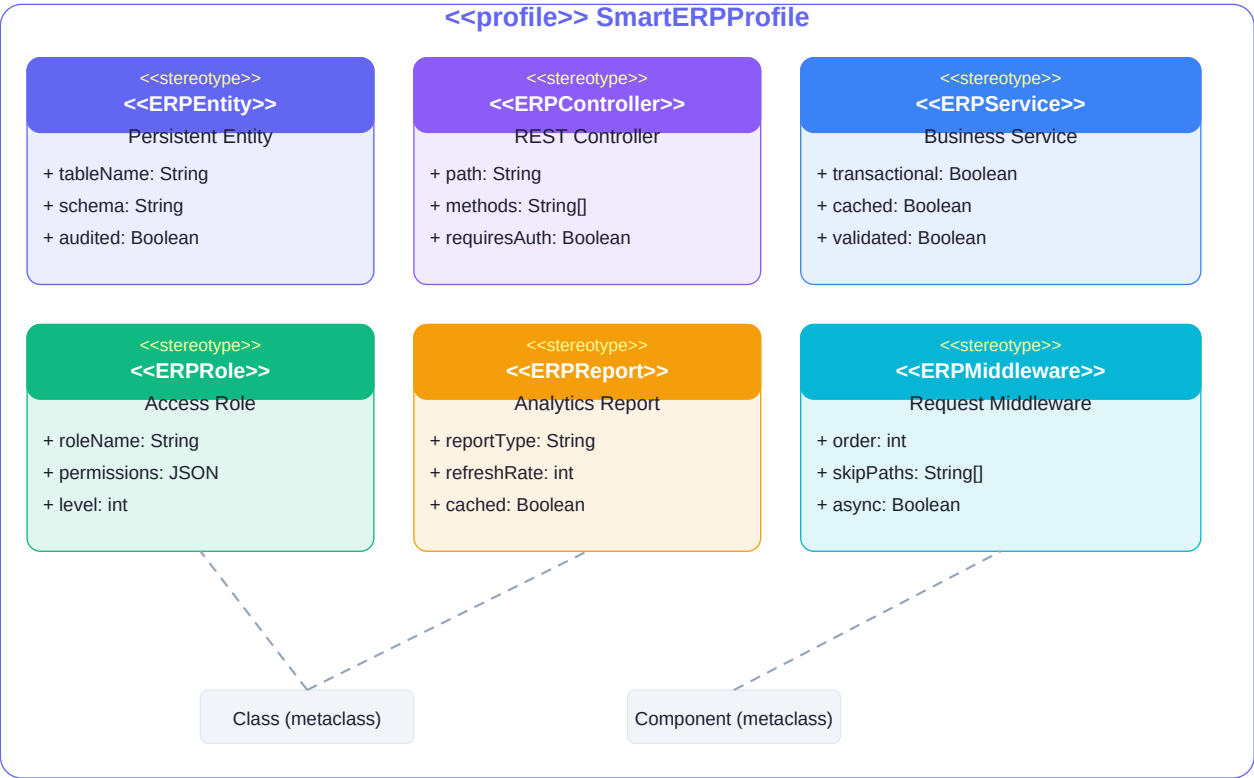


Figure 7: Profile Diagram defining custom stereotypes for the Smart ERP domain model

Stereotype	Extends	Tagged Values	Applied To
<<ERPEntity>>	Class	tableName, schema, audited	Student, Teacher, Course, etc.
<<ERPController>>	Component	path, methods, requiresAuth	All Controller classes
<<ERPService>>	Component	transactional, cached, validated	All Service classes
<<ERPRole>>	Class	roleName, permissions, level	Admin, Teacher, Student roles
<<ERPReport>>	Component	reportType, refreshRate, cached	Analytics, Dashboard
<<ERPMiddleware>>	Component	order, skipPaths, async	Auth, CORS, Logging

8. Use Case Diagram

UML Behavioral — System actors and use cases

Shows the functional requirements of the system from the user's perspective, identifying actors and their interactions with the system.

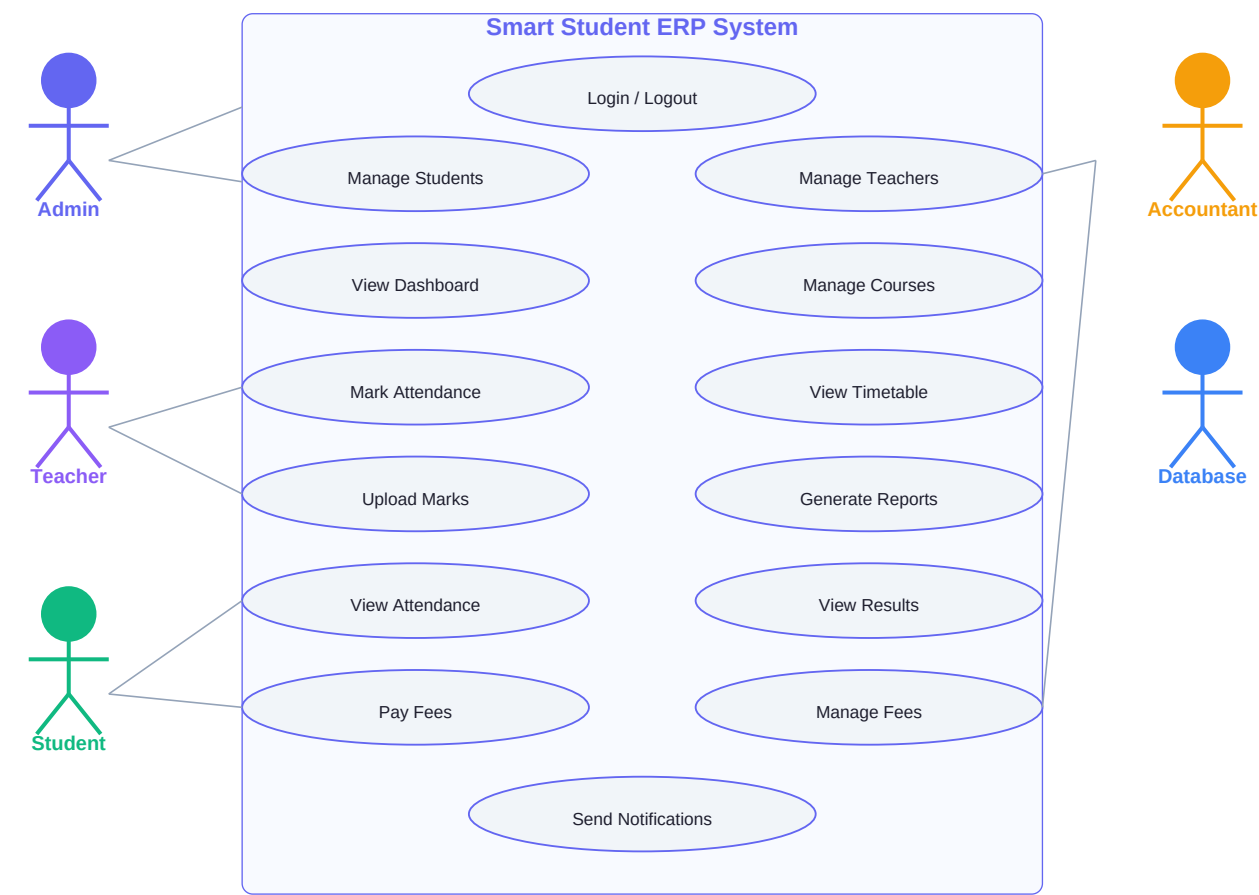


Figure 8: Use Case Diagram — all system actors and their primary use cases

Actor	Primary Use Cases	Permissions
Admin	Manage Students, Teachers, Courses; Send Notifications; View Reports	Full system access
Teacher	Mark Attendance, Upload Marks, View Timetable, Manage Courses	Academic data write access
Student	View Attendance, Results, Timetable, Fees; Library access	Read-only own data
Accountant	Manage Fees, Record Payments, Generate Invoices, Fee Reports	Financial data access
System	Auto-grade calculation, Notification broadcast, Late fine calculation	Background automated tasks

9. Activity Diagram

UML Behavioral — Student Registration Process

Shows the workflow/flow of control for the student registration process, including decision points and parallel activities.

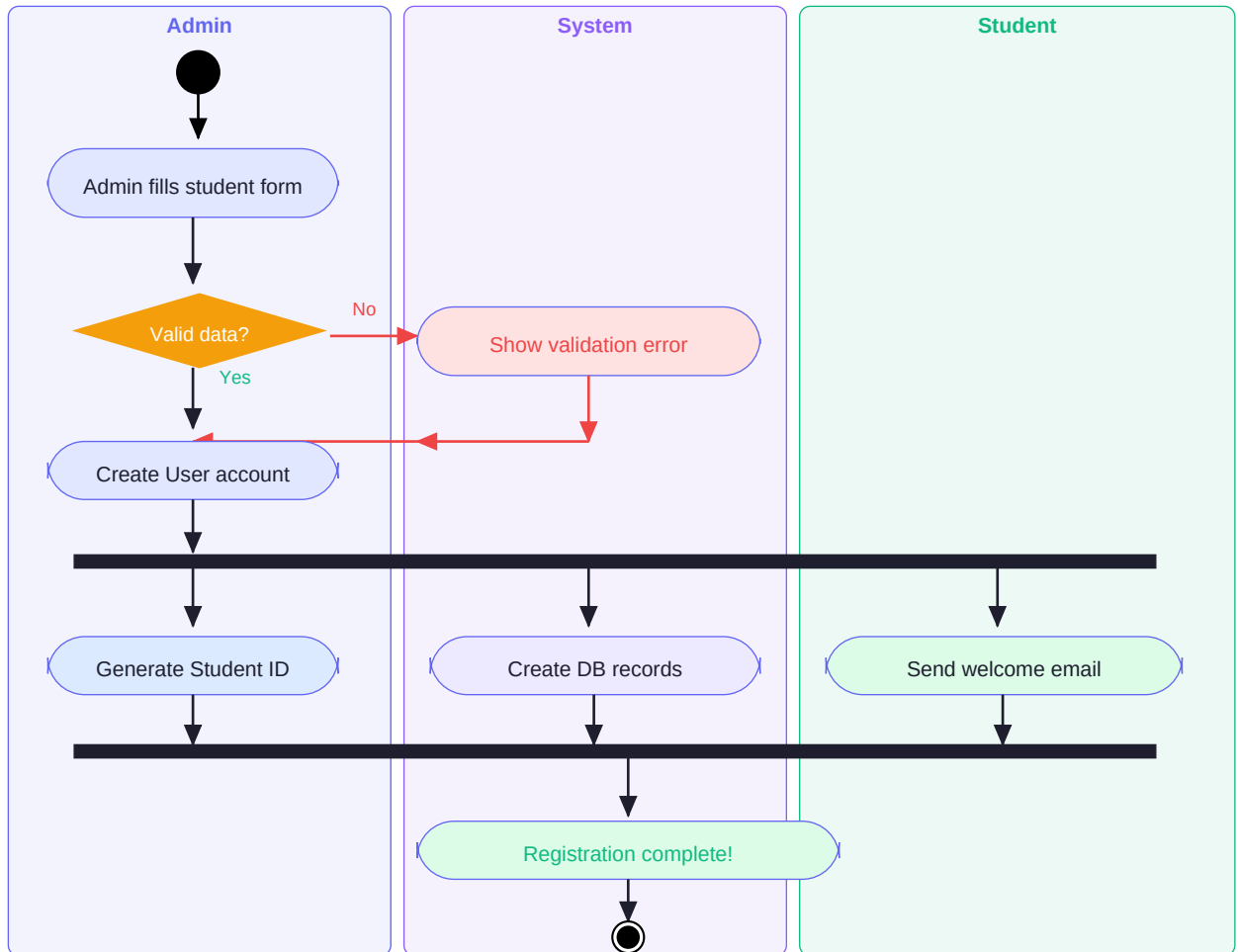


Figure 9: Activity Diagram — parallel student registration process across Admin, System, and Student swimlanes

10. State Machine Diagram

UML Behavioral — Student Enrollment Lifecycle

Shows the states of a Student object and the transitions between them based on events/triggers.

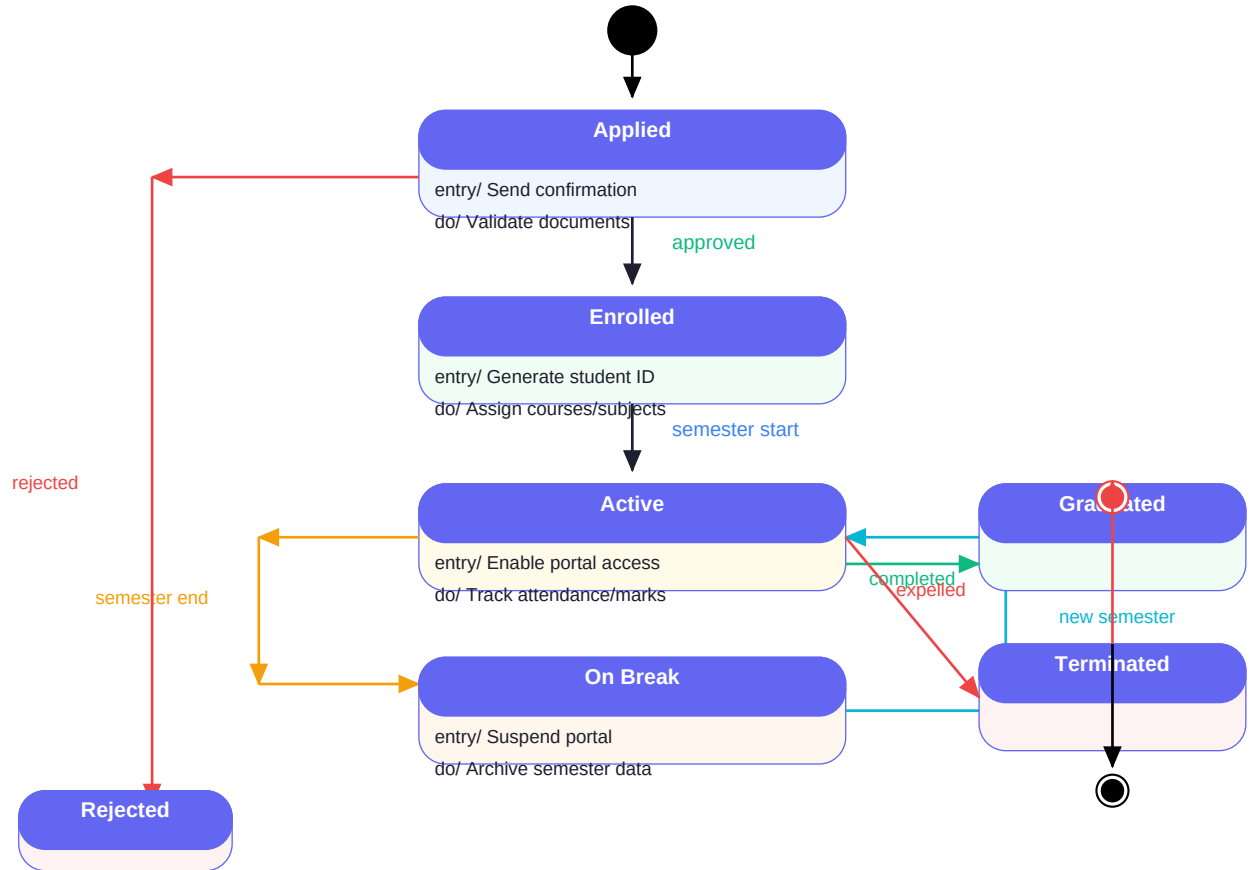


Figure 10: State Machine Diagram — complete Student enrollment lifecycle states and transitions

State	Entry Action	Do Activity	Exit Condition
Applied	Send confirmation email	Validate documents	Admin approves/rejects
Enrolled	Generate Student ID	Assign subjects/courses	Semester begins
Active	Enable portal access	Track attendance, marks	Semester ends
On Break	Suspend portal access	Archive semester data	New semester begins
Graduated	Generate certificate	Final transcript creation	Final end state
Terminated	Disable all access	Archive all records	Final end state
Rejected	Send rejection email	—	Appeal possible

11. ER Diagram (Entity-Relationship)

Database Design — Core entities and relationships

The Entity-Relationship Diagram shows all database entities, their attributes, and relationships including cardinality (1:1, 1:N, M:N).

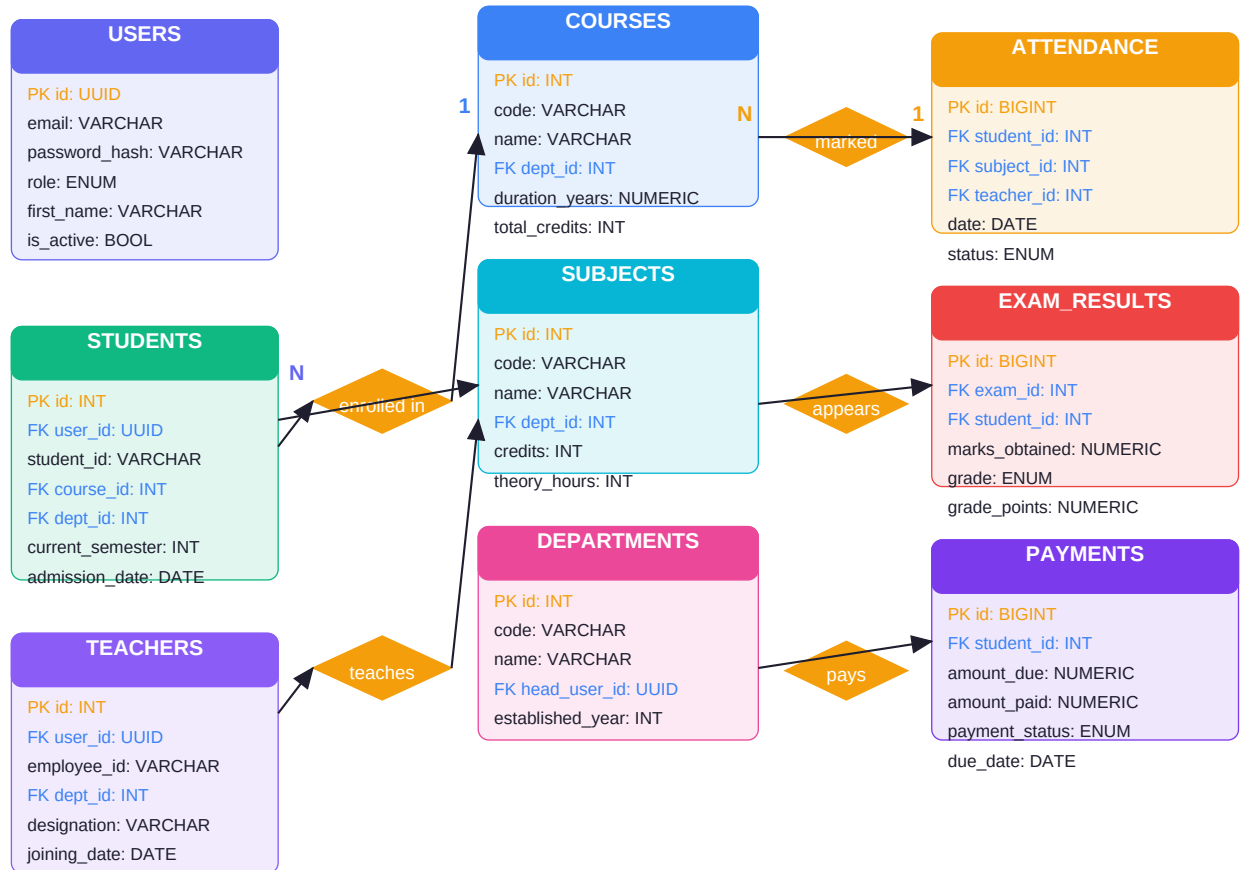


Figure 11: ER Diagram — complete database entity relationships with cardinality notation

Color Legend: Yellow=Primary Key, Blue=Foreign Key. Relationships: Students ENROLL in Courses (M:N), Teachers TEACH Subjects (M:N), Students have Attendance Records (1:N), Students have Exam Results (1:N), Students make Payments (1:N).

12. EER Diagram (Enhanced ER)

Database — Specialization, Generalization & Inheritance

EER extends ER with specialization/generalization, categories, and inheritance constraints showing how User is a superclass of Student, Teacher, and Accountant.

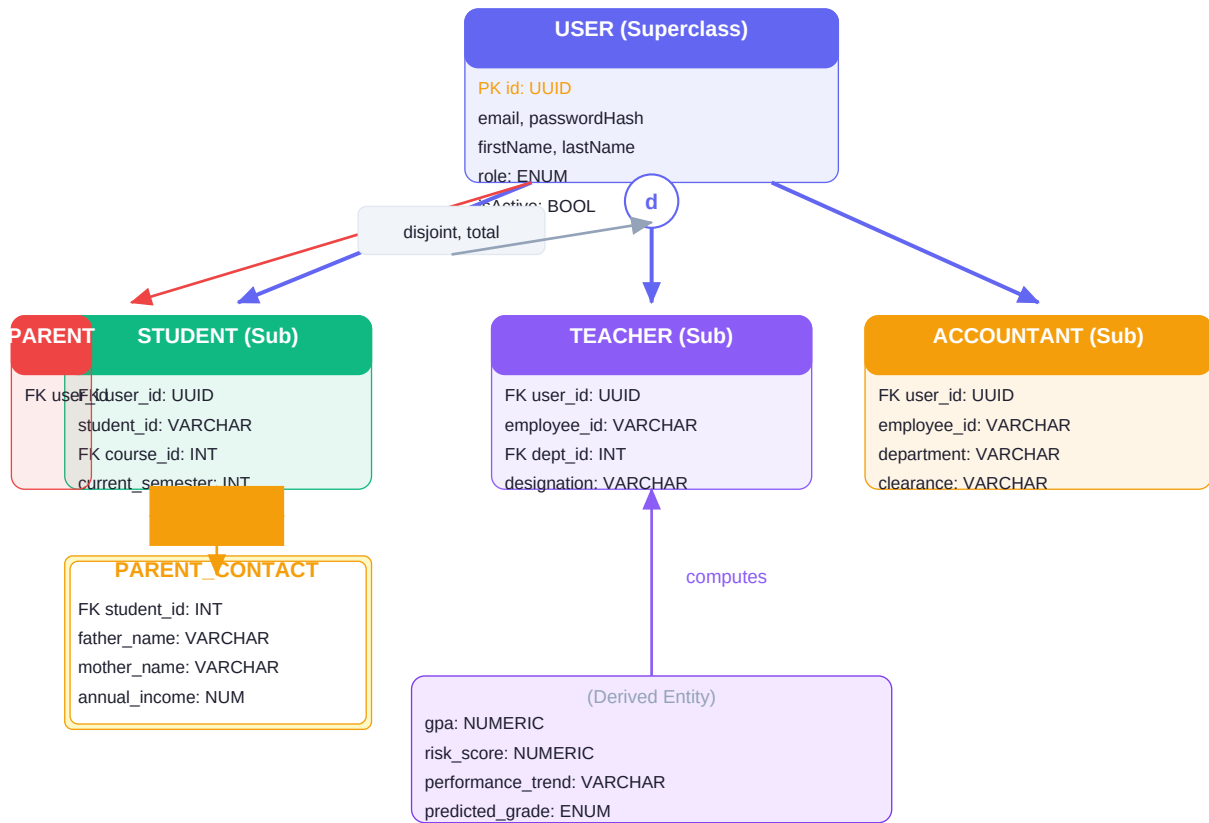


Figure 12: EER Diagram showing User superclass with disjoint total specialization into Student, Teacher, Accountant

EER Concept	Applied To	Constraint	Description
Superclass	USER entity	—	Base entity with common attributes for all users
Specialization	Student, Teacher, Accountant	Disjoint, Total	Each user IS EXACTLY ONE of the subclasses
Inheritance	All subclasses	—	Subclasses inherit id, email, name from USER
Weak Entity	PARENT_CONTACT	Identifying	Depends on STUDENT for its identity
Derived Entity	PERFORMANCE_ANALYTICS	—	Values computed from other entities (GPA, risk score)
Category	PARENT (optional)	Partial	Some users may optionally be parents

13. Relational Schema Diagram

Database — Normalized table schemas with keys

Shows the complete relational database schema with all tables, primary keys, foreign keys, and normalization to 3NF.

Schema.Table	Primary Key	Foreign Keys	Key Attributes	Constraints
auth.users	id (UUID)	—	email, password_hash, role, first_name, last_name	UNIQUE(email), role IN enum
auth.sessions	id (UUID)	user_id → users	token, ip_address, expires_at	UNIQUE(token), expires_at > NOW()
auth.login_logs	id (BIGINT)	user_id → users	ip_address, user_agent, status, logged_at	status IN (success, failed)
academic.departments	id (INT)	head_user_id → users	code, name, established_year, is_active	UNIQUE(code)
academic.courses	id (INT)	department_id → departments	code, name, duration_years, total_credits	UNIQUE(code), duration_years > 0
academic.subjects	id (INT)	department_id → departments	code, name, credits, theory_hours, lab_hours	UNIQUE(code), credits > 0
academic.students	id (INT)	user_id → users course_id → courses department_id → departments	student_id, current_semester, admission_date, enrollment_group	UNIQUE(student_id), UNIQUE(user_id)
academic.teachers	id (INT)	user_id → users department_id → departments	employee_id, designation, qualification, joining_date	UNIQUE(employee_id), UNIQUE(user_id)
academic.attendance	id (BIGINT)	student_id → students subject_id → subjects teacher_id → teachers	date, status, remarks, qr_token	UNIQUE(student_id,subject_id,date)
academic.exams	id (INT)	subject_id → subjects created_by → users	name, exam_type, exam_date, total_marks, passing_marks	UNIQUE(name), UPDATE (for new)
academic.exam_results	id (BIGINT)	exam_id → exams student_id → students	marks_obtained, grade, grade_points, is_absent	UNIQUE(exam_id,student_id), AUTO-GRADE
finance.payments	id (BIGINT)	student_id → students fee_structure_id → fee_structures	payment_ref, amount_due, amount_paid, payment_date	UNIQUE(payment_ref), AUTO late_fine trigger
finance.invoices	id (BIGINT)	payment_id → payments student_id → students	invoice_number, invoice_date, total_amount	UNIQUE(invoice_number), AUTO-GENERATE
library.books	id (INT)	subject_id → subjects	isbn, title, author, publisher, total_copies, available_copies	UNIQUE(isbn), available_copies >= 0
library.borrow_records	id (BIGINT)	book_id → books user_id → users	borrowed_at, due_date, returned_at, late_fine	AUTO late_fine trigger on return
communication.notifications	id (BIGINT)	created_by → users target_user_id → users	title, message, type, target_role, is_broadcast	AUTO-BROADCAST trigger on insert

Figure 13: Complete Relational Schema — 17 tables across 5 PostgreSQL schemas in 3NF

Normalization: All tables are in Third Normal Form (3NF). No transitive dependencies. Each non-key attribute depends only on the primary key. Schemas: auth (4 tables), academic (10 tables), finance (3 tables), library (2 tables), communication (2 tables).

14. Data Flow Diagram — Level 0 (Context)

DFD — System context showing external entities

Level 0 DFD (Context Diagram) shows the entire system as a single process with all external entities and major data flows.

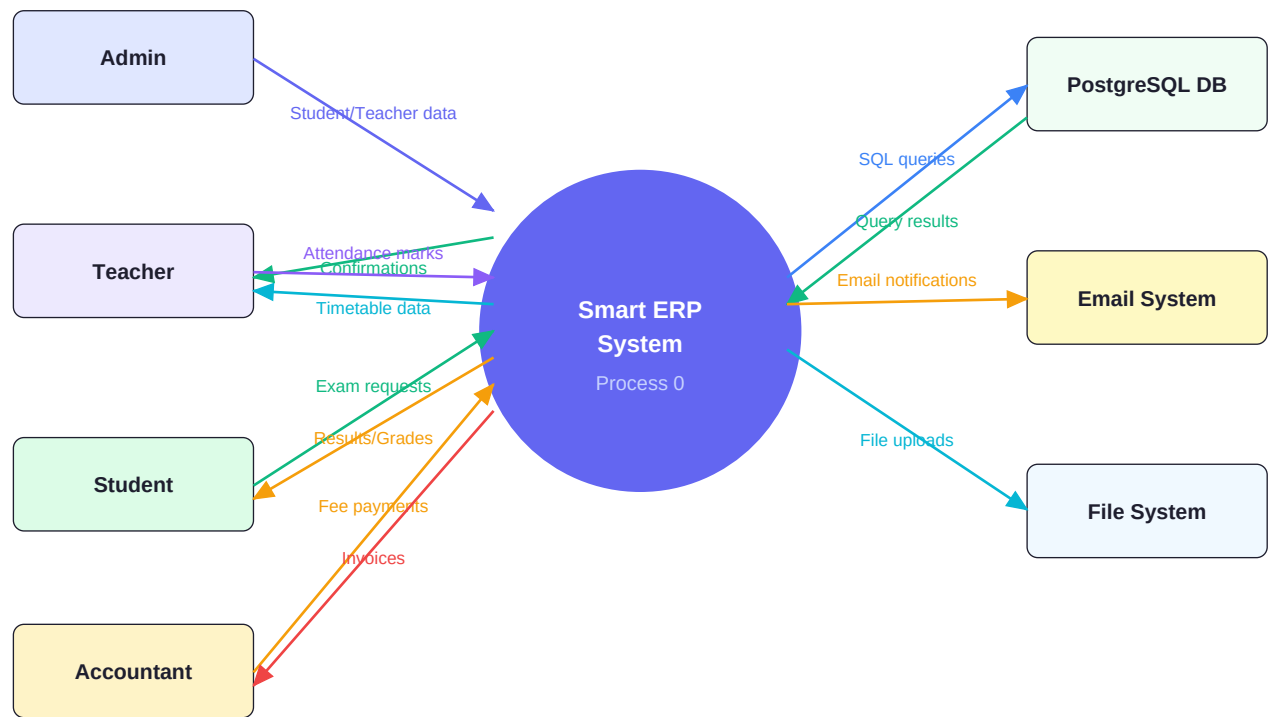


Figure 14: Level 0 DFD — Context Diagram showing Smart ERP as single system with all external entities

External Entity	Data Flow To System	Data Flow From System
Admin	Student registrations, teacher data, course setup, announcements	Dashboard reports, confirmations, system alerts
Teacher	Attendance marks, exam results, timetable requests	Student lists, timetable, attendance summaries
Student	Fee payments, library requests, profile updates	Results, timetable, attendance record, fee invoices
Accountant	Fee structure setup, payment records	Payment reports, outstanding fee lists, invoices
PostgreSQL DB	—	Stored/retrieved all persistent data
Email System	—	OTP emails, notifications, fee reminders

15. Data Flow Diagram — Level 1

DFD — Expanded sub-processes within Smart ERP

Level 1 DFD decomposes the central process into major sub-processes showing detailed data flows between modules.

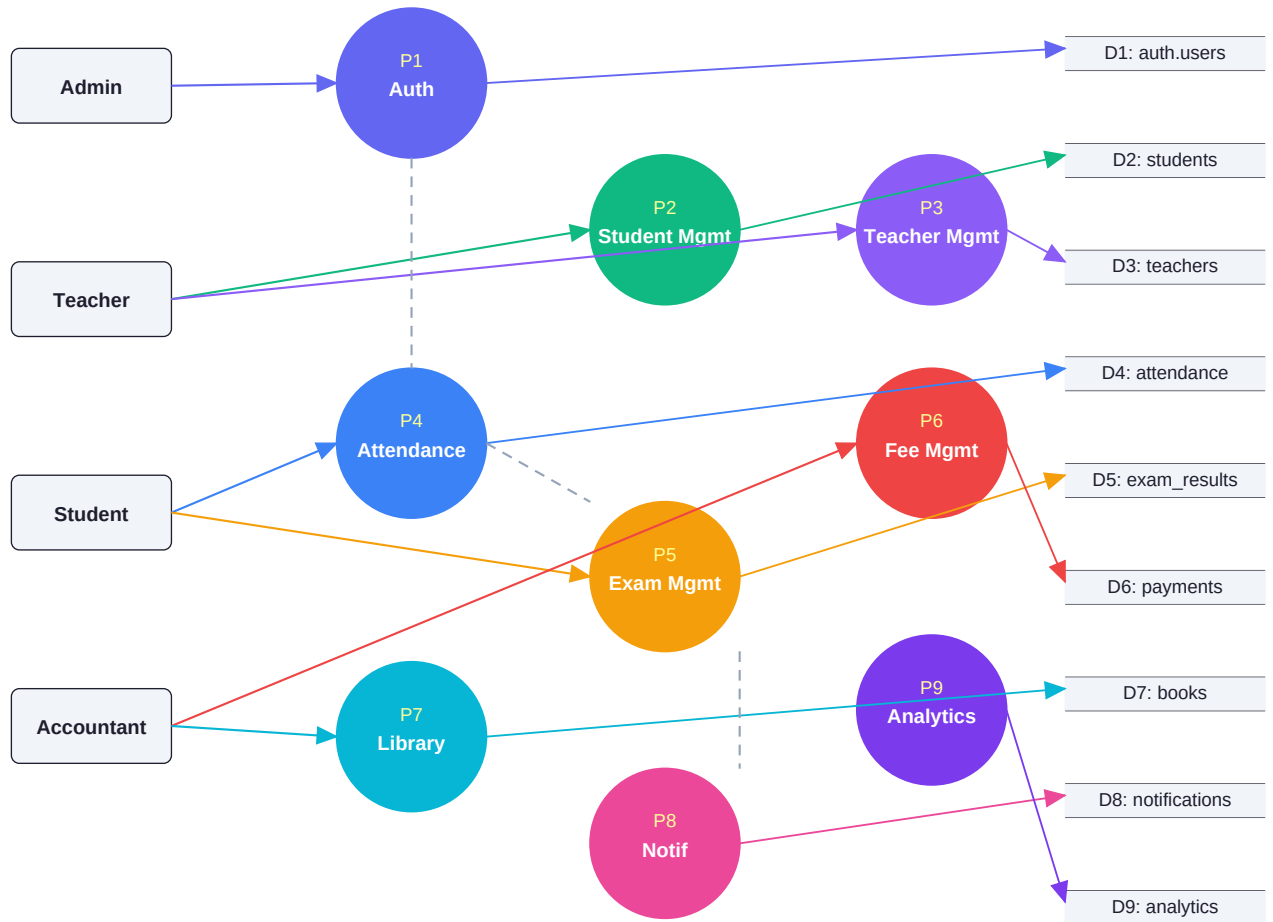


Figure 15: Level 1 DFD — all 9 sub-processes (Auth, Student, Teacher, Attendance, Exam, Fee, Library, Notifications, Analytics) with data stores

16. Flowchart — Login & Authentication Process

Flowchart — Step-by-step login decision logic

Detailed flowchart of the login process including OTP verification, role-based routing, and session management.

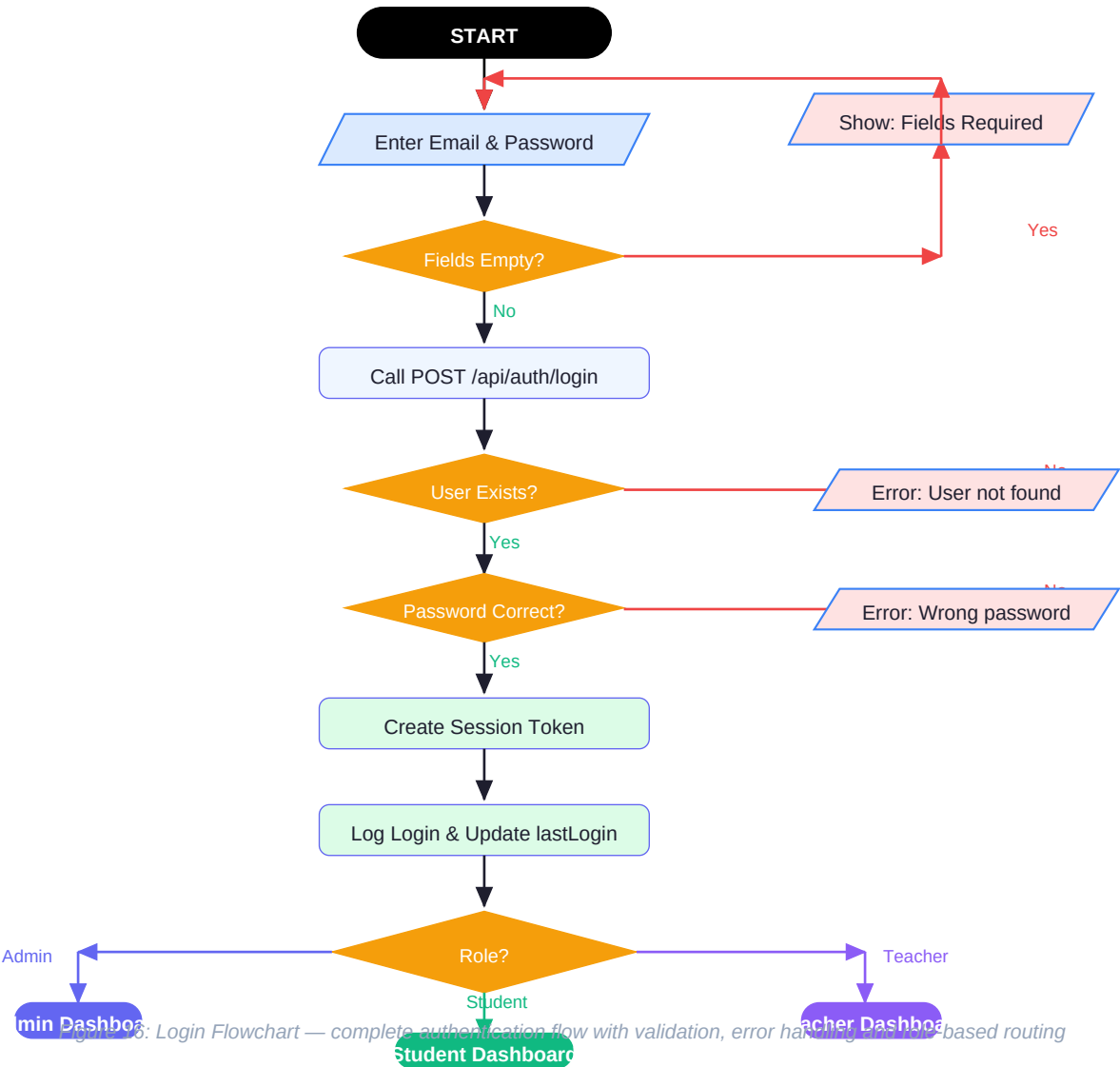


Figure 16: Login Flowchart — complete authentication flow with validation, error handling, and role-based routing

17. Flowchart — Fee Payment Process

Flowchart — End-to-end fee payment workflow

Shows the complete fee payment process from student initiating payment to invoice generation.

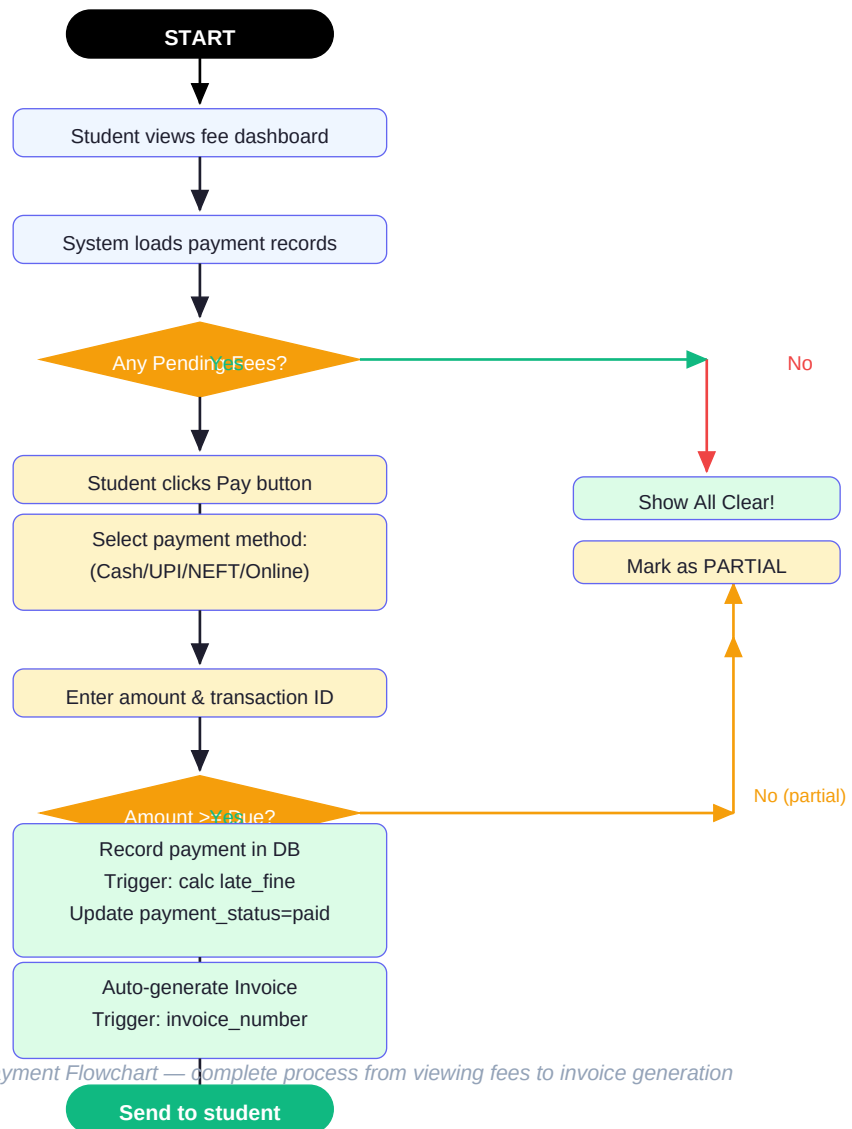


Figure 17: Fee Payment Flowchart — complete process from viewing fees to invoice generation

18. Workflow Diagram — Academic Year Process

Workflow — Complete academic year management workflow

Shows the end-to-end workflow for managing a complete academic year in the Smart ERP system, from setup to completion.



Figure 18: Workflow Diagram — complete academic year management across 5 phases

Phase	Duration	Key Activities	Stakeholders
1. Setup Phase	Aug (Week 1-2)	Academic year config, fee setup, timetable creation, room allocation	Admin
2. Enrollment Phase	Aug (Week 3-4)	Student registration, ID generation, subject assignment, notifications	Admin, Student
3. Academic Phase	Sep-Apr (8 months)	Daily attendance, mid/final exams, mark upload, progress tracking	Teacher, Student
4. Financial Phase	Throughout year	Fee collection, invoice generation, scholarship, defaulter tracking	Accountant, Student
5. Completion Phase	May-Jul (3 months)	GPA calculation, result publishing, student promotion, data archiving	Admin, All